

Distribution of Compounded Real Returns over 30 Years*Distribution of Compounded Real Returns over 30 Years**Monte Carlo Simulations for a 50/50 Asset Allocation**Based on SBBI Data, 1926–2016, S&P 500 and Intermediate-Term Government Bonds*

	Lump Sum	Accumulation	Retirement
1st Percentile	0.7%	0.2%	–0.1%
5th Percentile	2.0%	1.7%	1.3%
10th Percentile	2.7%	2.4%	2.1%
25th Percentile	3.8%	3.7%	3.4%
Median	5.1%	5.2%	4.9%
75th Percentile	6.4%	6.6%	6.6%
90th Percentile	7.6%	7.9%	8.3%
95th Percentile	8.3%	8.7%	9.3%
99th Percentile	9.7%	10.2%	11.4%
Mean	5.1%	5.2%	5.1%
Std. Deviation	1.9%	2.1%	2.5%

Source: Own calculations with one hundred thousand Monte Carlo simulations for thirty-year periods. Portfolio returns are lognormally distributed with a 5.6 percent arithmetic real return and 10.7 percent volatility.

In all three cases, the median return was close to 5.1 percent, which was the assumed compounded return input (given the 5.6 percent arithmetic return and 10.7 percent volatility) for the simulations. However, after thinking about it, most retirees would probably shy away from using this number to develop a financial plan with a fixed return in a spreadsheet. The probability the median return can be achieved is only 50 percent. When choosing a number to plug into a spreadsheet, a conservative retiree might be more comfortable using something like the return in the twenty-fifth percentile—or even the tenth—of the distribution. These numbers would correspond to 75 or 90 percent probabilities of success, respectively, for a financial plan created with planning software. I worry that this point is too often forgotten. People tend to think in terms of an average return rather than a more conservative return that would provide a better chance for the plan to work.